

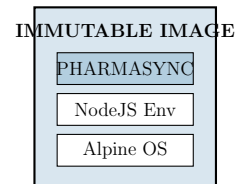
PharmaSync V4: Technical Defense

Core Systems Architecture Presentation Cards

CARD 1: DOCKER - IMMUTABLE INFRASTRUCTURE

[CONCEPT] "We ship a complete, pre-configured server environment."

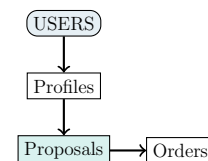
- **Containerization:** Isolated userspace ensures the app behaves identically on any OS.
- **Alpine Base:** Built on a security-hardened, minimalist Linux distribution to reduce attack surfaces.
- **Auto-Bootstrap:** Detects empty volumes and executes the SQL schema automatically during the "Entry point" phase.



CARD 2: RELATIONAL CORE & INTEGRITY

[CONCEPT] "Highly normalized data structure with strict referential integrity."

- **3rd Normal Form (3NF):** Eliminates data redundancy and ensures logical consistency across actors.
- **Enforced Logic:** Uses ON DELETE CASCADE to ensure medical records are purged when user accounts are closed.
- **Role Segregation:** Strict ENUM constraints enforce access control for Admins, Pharmacists, and Patients.



CARD 3: ATOMIC MARKET TRANSACTIONS

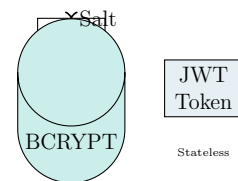
[CONCEPT] "Ensuring the 'ACID' properties of the clinical marketplace."

- **Transactional Atomicity:** Uses `beginTransaction()` so that restocks and payment logs succeed or fail as a single atomic unit.
- **Uber-Handshake:** Replaces static lists with a dynamic Bidding State Machine (Pending → Accepted/Declined).
- **Proximity Awareness:** System ranks providers based on distance to optimize clinical delivery time.

CARD 4: THE SECURITY FORTRESS

[CONCEPT] "Stateless, cryptographic defense of patient data."

- **Bcrypt Hashing:** Passwords are never stored. We store a 60-character salted hash with 10 work rounds to block brute-force attacks.
- **JWT Authentication:** Uses a Header-Payload-Signature structure. This enables stateless verification without needing server-side session memory.



CARD 5: SMART UX & NATIVE NAVIGATION

[CONCEPT] "Making a web app behave like a native desktop tool."

- **History API:** Directly manipulates the browser's `pushState` stack to enable Back/Forward buttons without reloads.
- **Graceful Fallback:** Automated SSL detection logic switches between HTTPS and HTTP modes to ensure maximum server uptime.
- **Vector Visuals:** Zero-image SVG architecture ensures professional clarity on any clinical display.

